

Using the Complement

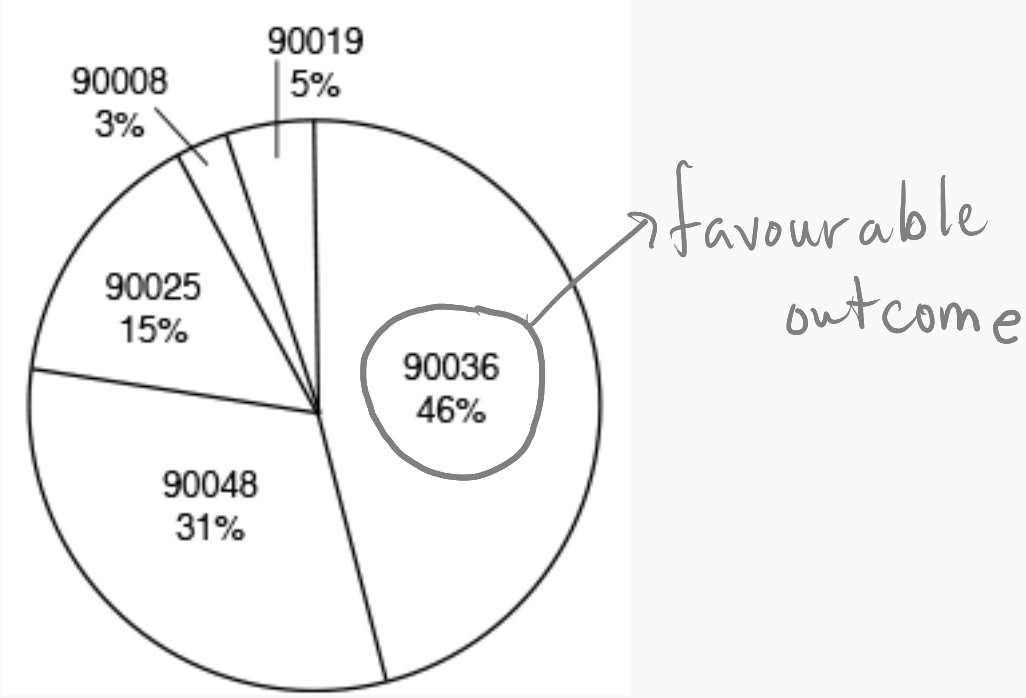
When given a set of collected data, you can use your knowledge of probability to draw conclusions about what will likely happen.

Example 1

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Rob’s Roofing Company offers area homeowners a free estimate for a new roof. The company tracks which zip codes yield positive responses to the special offer. The information is shown on the graph below.

Percent of Positive Responses to Free Offer by Zip Code



From the names of the homeowners who accepted a free estimate, one name is chosen randomly to win a prize. What is the probability that the name will be from the 90036 zip code area?

The graph represents all positive responses to the promotion. The highest percent on the graph is 46%, so there is a 46% chance of drawing a name from zip code 90036.

The probability is 46%.

Example 2

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What is the chance that the name will *not* be from the 90036 zip code?

Since there is a 100% chance that the name will either be in the zip code or not, the two situations are complements. Because of this, the first probability can be subtracted from 100% to get its complement.

Subtract: $100\% - 46\% = 54\%$

Since 54% is greater than 46%, it is more likely that the name will be drawn from a zip code other than 90036.

The chance is **54%**.

Example 3

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If the employees make 220 calls to the 90025 zip code, how many positive responses will they probably get?

You can reasonably conclude that Rob’s Roofing Company will continue to have 15% success in the 90025 zip code.

Find 15% of 220.

$220 \times 0.15 = 33$

Of the 220 calls, **33 responses** will probably be positive.

"of" means multiply
example: 20% of 30 = $(0.20)(30) = 6$

Think about Examples 1 and 2. If someone asked you which zip code the winner would be from, you would probably say 90036. It has the greatest percent. But you would have a better chance of being right if you said the zip code *would not be* 90036.

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Lesson 5: Probability

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